ARYAN MEENA

Quantitative Analyst | Python | SQL | Excel | Risk Modeling

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SKILLS

Python, SQL, Excel, pandas, NumPy, statistical analysis, hypothesis testing, regression, time series analysis, backtesting, risk modeling, VaR, Expected Shortfall, GARCH, Monte Carlo simulation, bootstrap, multiple-testing correction, data cleaning, data validation, Bloomberg data, Power BI, Tableau, PostgreSQL, AWS, Git

EXPERIENCE

Boston University | ***Quantitative Researcher (Research Assistantship)*** Sep 2025 – May 2026

- Traced a 15-year trading signal to biased opening prices in a free data feed (Sharpe -1.36 on vendor data vs +0.25 on exchange data), protecting the research from a false result; paper **published in Risks (MDPI)**.
- Designed a carry-screened long-only commodity index that beat the Bloomberg Commodity Index by **5.3% a year** (2008–2026, information ratio 0.70, t 3.33) with one rebalance a year and a shallower drawdown (-53% vs -75%).
- Replicated a published tail-skewness premium point in time: Sharpe 0.65 (t 3.46), the 1 of 8 pre-registered factors to survive Holm correction, and stronger after publication (0.82 vs 0.50).
- Modeled per-contract trading costs and capacity (spreads, rolls, square-root market impact), showing a $100M fund keeps a net Sharpe of 0.59 with the largest position near 0.7% of open interest.
- Showed near-zero correlation with stocks, Treasuries and commodities (-0.01, -0.05, 0.07) and that a quarter-sized overlay doubles the commodity index's Sharpe ratio (0.15 to 0.31).
- Stress-tested the strategy through 4 market crises, gaining 14.8% in late 2008 while BCOM fell 49%.
- Compared momentum, median and contrarian rotation across 15 commodity futures and 6 ETFs; monthly ETF winners grew $100 to $337 vs $121 for equal weight (paper under review, Commodities).
- Rewrote the backtest engine in NumPy to run 1,312 strategy configurations 17x faster (1.0 s vs 17.3 s) with identical results, making 2,000 bootstrap stress tests practical on just two CPU cores.

Funding Pips (Remote) | ***FX Trader, Funded Account*** Dec 2023 – Dec 2024

- Hit a profit target of at least 15% of account size 3 times in a row to earn release of real trading capital on a funded prop-firm account, never breaching the 10% maximum or 5% daily drawdown limit.
- Traded major currency pairs intraday for 12 months under hard loss limits, learning the position sizing and risk controls that later shaped how the BU research tested and validated every trading strategy.

PROJECTS

Multi-Asset Risk Engine | Python, VaR/ES, GARCH, pytest GitHub

- Measured risk in a multi-asset portfolio, tracing 52% of the risk to a 40% equity weight and showing a normal model understated 99% tail loss by about a third (11.1% vs 7.4% expected shortfall).
- Backtested 8 Value-at-Risk models over 2,796 days and selected the 1 that met bank standards with ~13% less VaR.

Service Operations Analytics | Python, SQL, AWS, GitHub Actions Live Dashboard

- Automated a daily AWS pipeline over 130K+ records that publishes only after 17 data checks pass, at about $0.03 per run, with alarms for broken or stale data so reports stay trustworthy.

EDUCATION

MS, Applied Data Analytics, Boston University (GPA 3.75/4.0) May 2026

- Led a six-person team at the BU MedAI Hackathon to a **podium finish** among 40+ teams; 5 first-author papers.
- Coursework: Data Mining, Data Analytics and Visualization, Foundations of Machine Learning, Big Data Analytics.

BTech, Electronics and Communication Engineering, Indian Institute of Space Science and Technology May 2023

- Coursework: Statistics and Probability, Computer Vision, Machine Learning for Signal Processing.